

A PROJECT TITLE
ON
CHRONIC KIDNEY DISEASE PREDICTION USING XGBOOST AND LIGHTGBM
GRADIENT-BOOSTED TREES

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF <DEPARTMENT NAME>

Name of Title: Chronic Kidney Disease Prediction Using XGBoost and LightGBM Gradient-Boosted Trees

1. Abstract:

Chronic Kidney Disease (CKD) is a progressive disorder that often remains undetected until it reaches an advanced stage, making early and accurate prediction essential for timely clinical intervention. This project presents a machine learning approach to predict the presence of CKD using two powerful gradient-boosted tree algorithms, XGBoost and LightGBM, and compares their predictive performance. The dataset used comprises clinical attributes such as blood pressure, blood glucose level, serum creatinine, haemoglobin, albumin, and other biochemical and haematological parameters commonly recorded during routine diagnosis. The raw data is first preprocessed through handling of missing values, encoding of categorical attributes, and normalization of numerical features to ensure consistency and improve model reliability. Exploratory data analysis is performed to understand feature correlations and identify the attributes most strongly associated with CKD onset. Both XGBoost and LightGBM models are then trained on the processed dataset, with hyperparameters tuned using techniques such as grid search and cross-validation to minimize overfitting and maximize generalization. The performance of the two models is evaluated and compared using standard classification metrics, including accuracy, precision, recall, F1-score, and the area under the ROC curve. Feature importance analysis is additionally carried out to identify the clinical parameters that contribute most significantly to the prediction, offering interpretability that can assist medical practitioners. Experimental results demonstrate that both ensemble models achieve high predictive accuracy, with LightGBM offering faster training time due to its histogram-based leaf-wise growth strategy, while XGBoost provides slightly more stable performance on smaller datasets. The proposed system can serve as a decision-support tool for early CKD screening, reducing diagnostic delay and enabling preventive healthcare measures. This work highlights the effectiveness of gradient-boosted ensemble methods in medical diagnosis and lays the foundation for future extensions involving larger datasets, deep learning hybrids, and deployment as a real-time clinical prediction system.

2. INTRODUCTION:

Chronic Kidney Disease is a major global health concern, affecting a significant proportion of the adult population and contributing heavily to morbidity and mortality worldwide. It is characterized by a gradual loss of kidney function over time, often progressing silently through its early stages without noticeable symptoms. By the time clinical symptoms become apparent, the disease has frequently advanced to a stage where treatment options are limited and costly, such as dialysis or transplantation. Early detection is therefore critical to slowing disease progression, reducing healthcare costs, and improving patient outcomes. With the increasing availability of electronic health records and clinical laboratory data, machine learning techniques have emerged as promising tools for building predictive models that can assist clinicians in early diagnosis. Among the various machine learning approaches, ensemble learning methods based on gradient boosting, such as XGBoost and LightGBM, have gained popularity due to their high predictive accuracy, robustness to noisy data, and ability to handle complex, non-linear relationships between features. This project focuses on leveraging these two gradient-boosted tree algorithms to build an effective CKD prediction system, comparing their performance to determine which approach offers a better trade-off between accuracy, training efficiency, and interpretability for this particular healthcare application.

3. SOFTWARE REQUIREMENTS:

- Operating System: Windows 10/11 or Linux (Ubuntu 20.04+)
- Programming Language: Python 3.9 or above
- Development Environment: Jupyter Notebook / Google Colab / VS Code
- Machine Learning Libraries: XGBoost, LightGBM, scikit-learn
- Data Handling Libraries: pandas, NumPy
- Visualization Libraries: Matplotlib, Seaborn
- Version Control: Git and GitHub



Signature of the Guide



Course Instructor